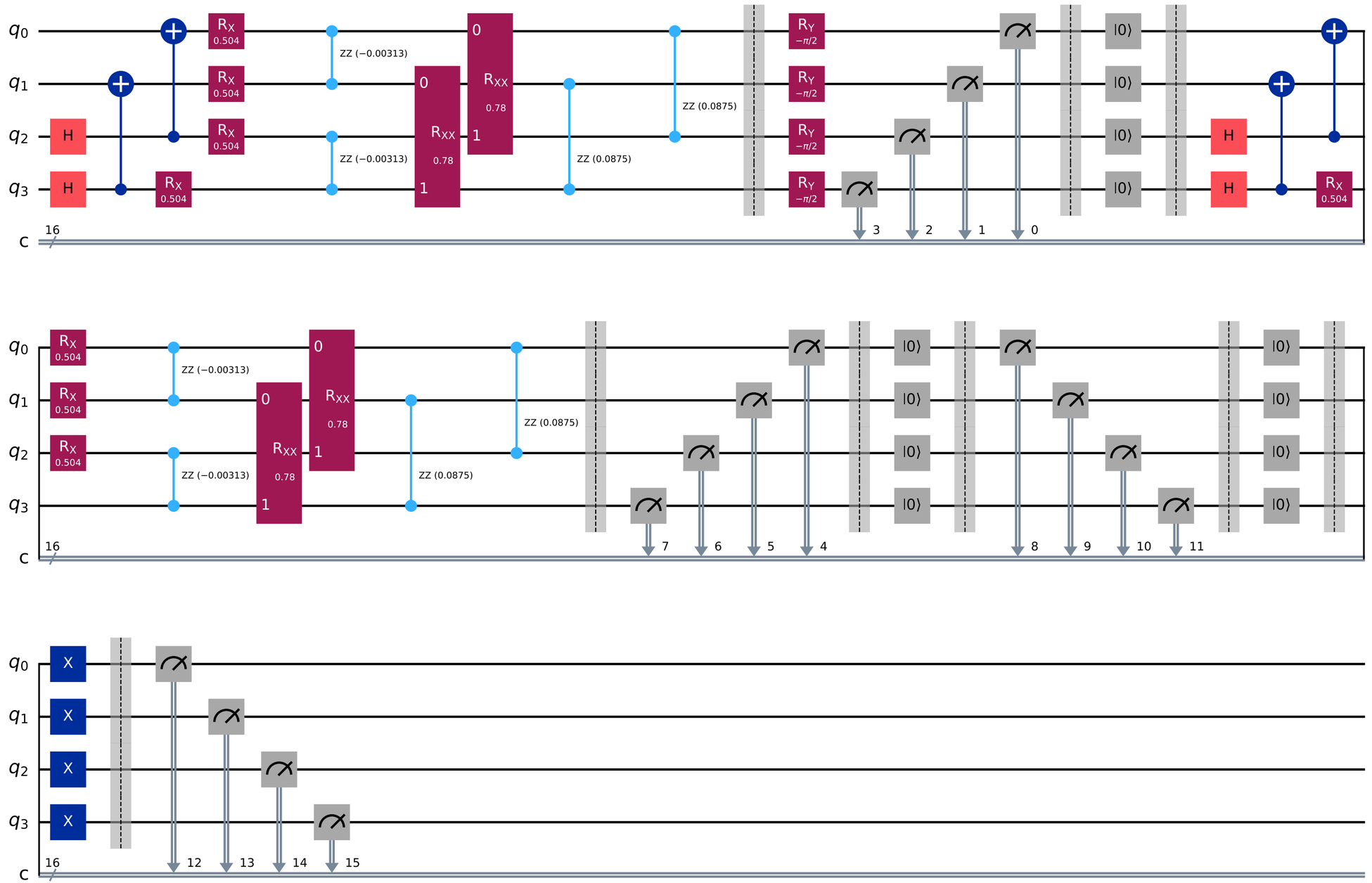
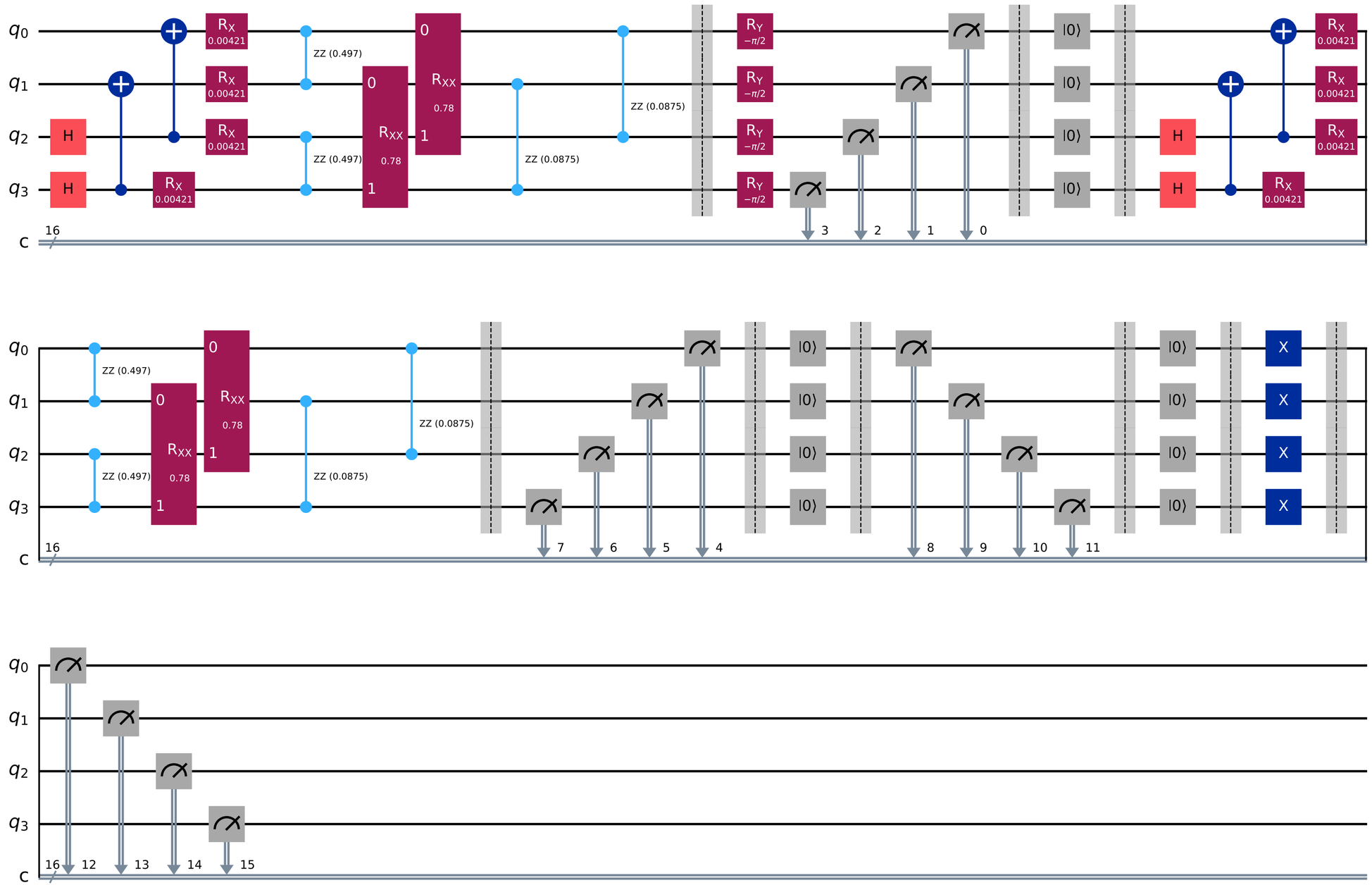
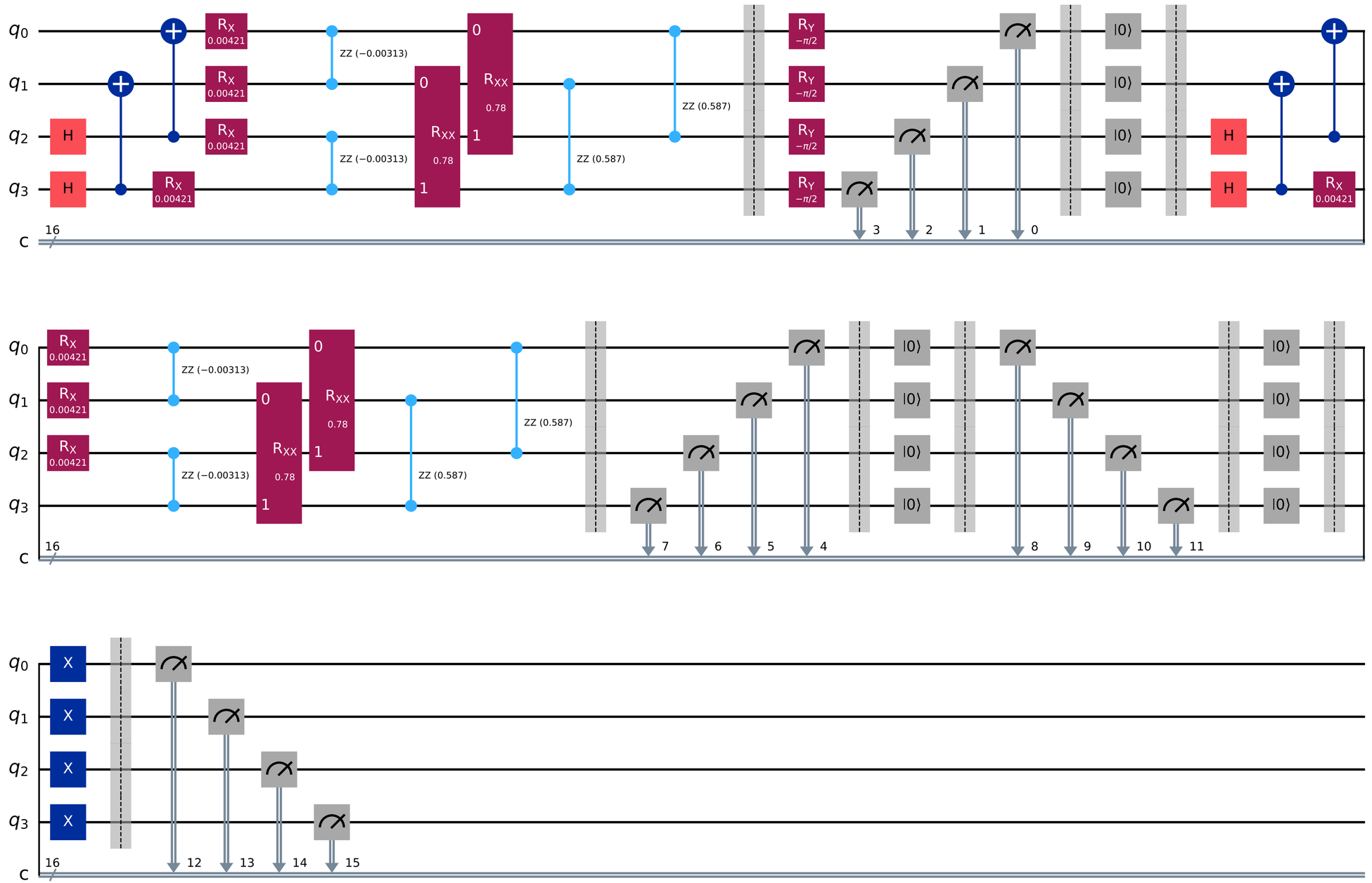
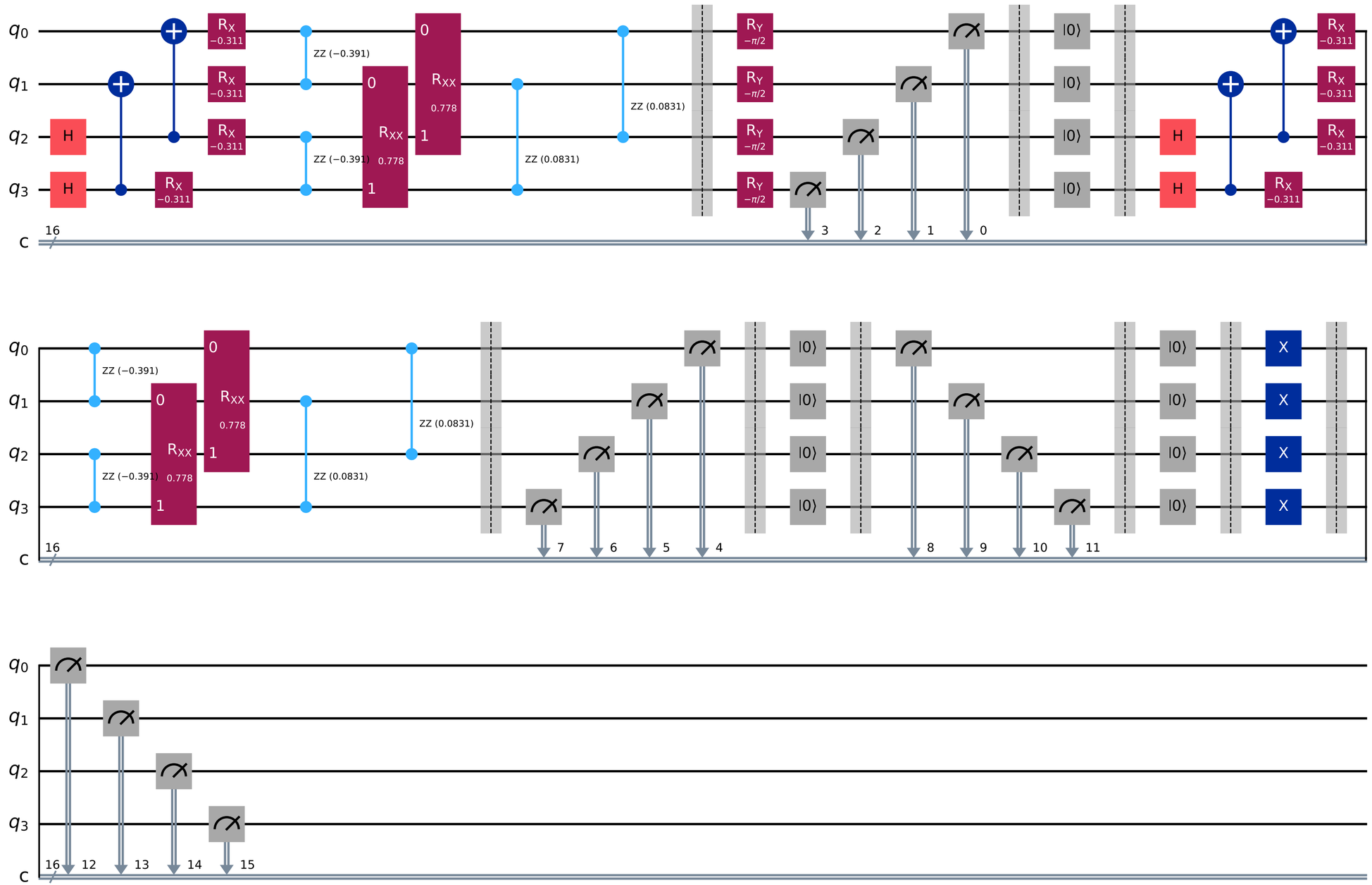
q0
q1
q2
q3
c
16
H
H
RX 0.00421
RX 0.00421
RX 0.00421
RX 0.00421
ZZ (−0.00313)
ZZ (−0.00313)
RXX 0.78
ZZ (0.0875)
ZZ (0.0875)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX 0.00421
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15
X
X
X
X

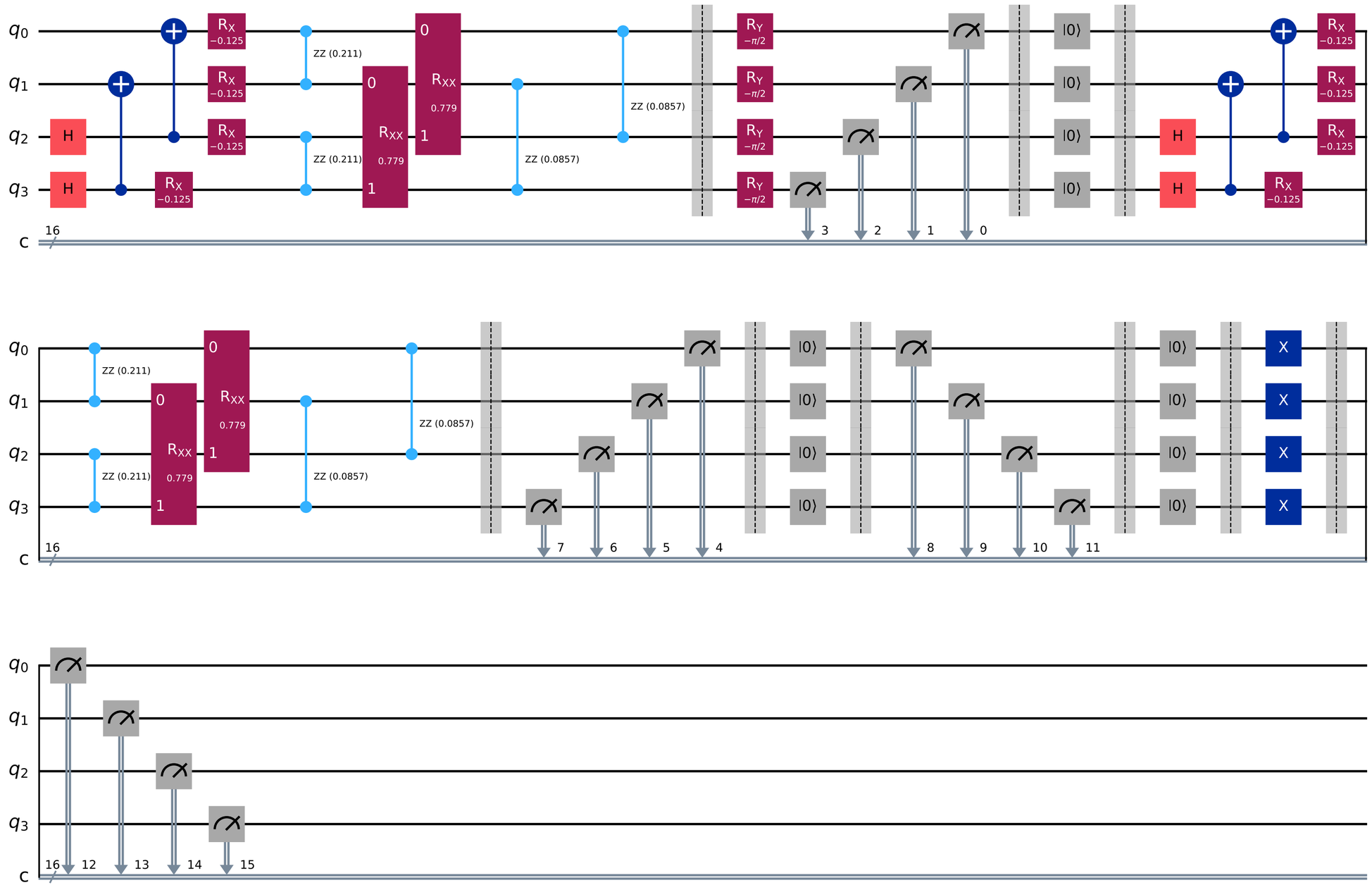

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.504
R_X 0.504
R_X 0.504
R_X 0.504
ZZ (−0.00313)
ZZ (−0.00313)
0
R_{XX} 0.78
1
0
R_{XX} 0.78
1
ZZ (0.0875)
ZZ (0.0875)
R_Y $-\pi/2$
R_Y $-\pi/2$
R_Y $-\pi/2$
R_Y $-\pi/2$
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X 0.504
3
2
1
0
R_X 0.504
R_X 0.504
R_X 0.504
ZZ (−0.00313)
ZZ (−0.00313)
0
R_{XX} 0.78
1
0
R_{XX} 0.78
1
ZZ (0.0875)
ZZ (0.0875)
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
X
X
X
X
12
13
14
15

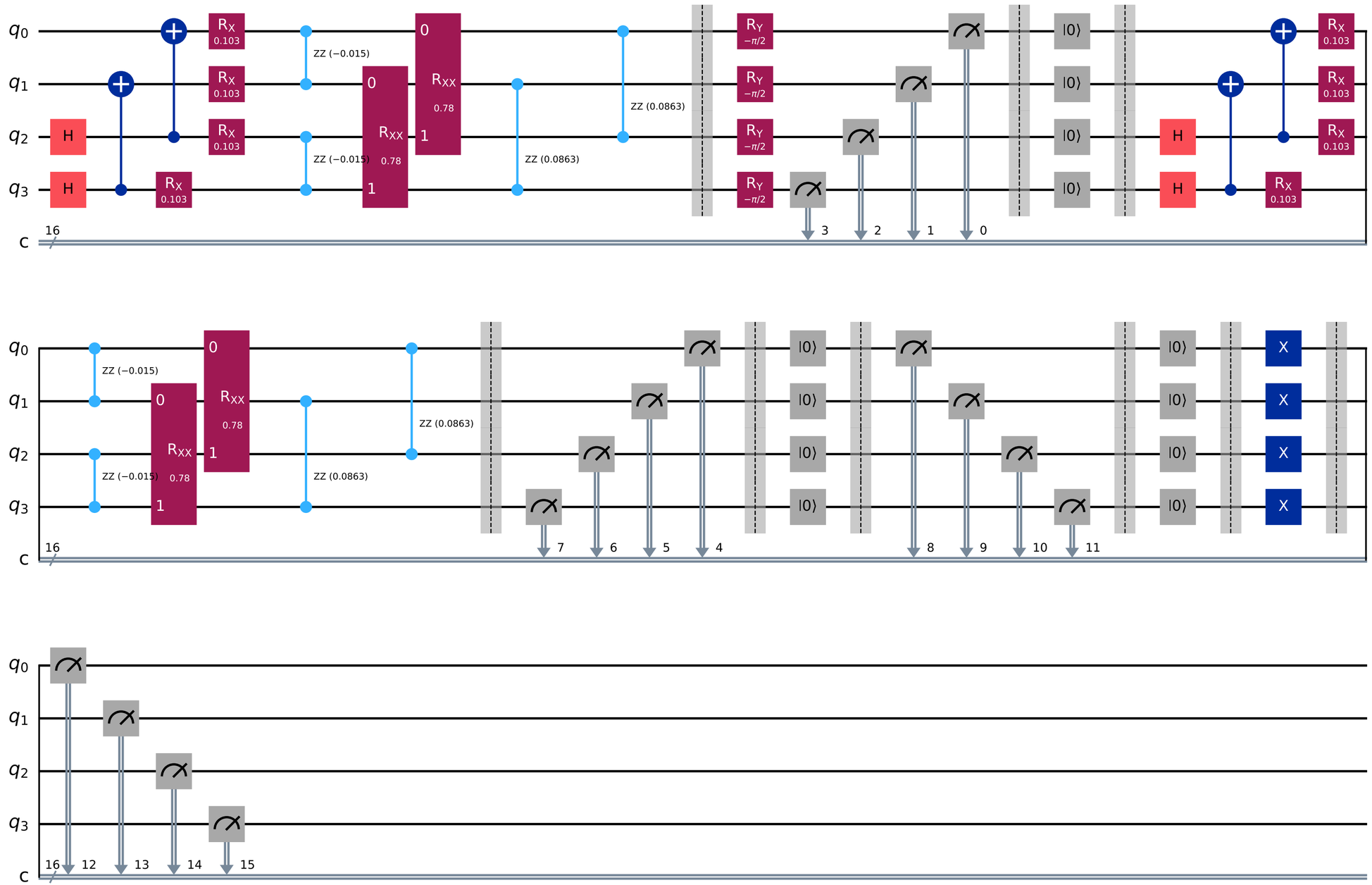

q0
q1
q2
q3
c
16
H
H
RX 0.00421
RX 0.00421
RX 0.00421
RX 0.00421
ZZ (0.497)
ZZ (0.497)
RXX 0.78
ZZ (0.0875)
ZZ (0.0875)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX 0.00421
RX 0.00421
RX 0.00421
RX 0.00421
ZZ (0.497)
ZZ (0.497)
RXX 0.78
RXX 0.78
ZZ (0.0875)
ZZ (0.0875)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
16
12
13
14
15

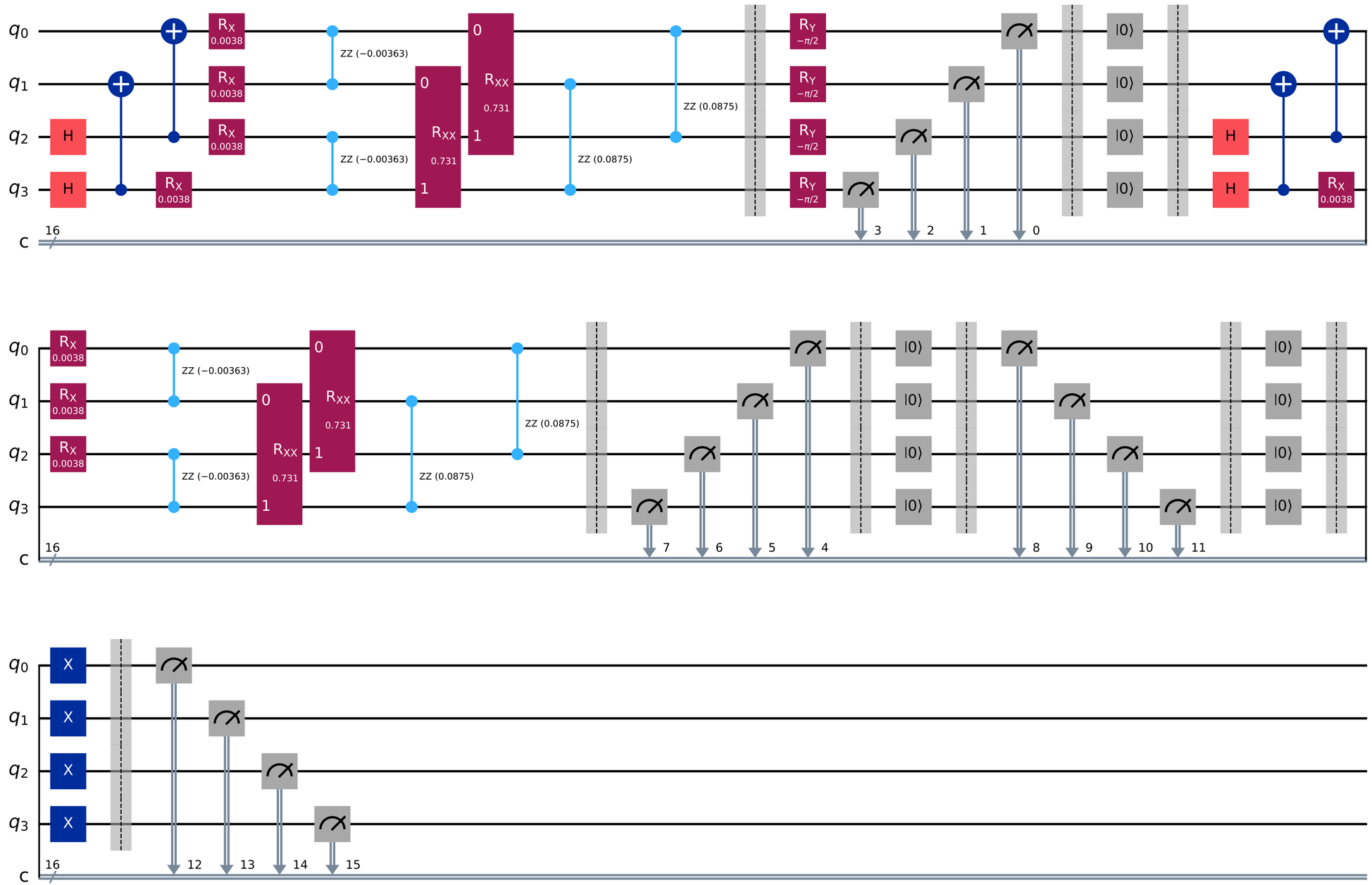
q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.00421
R_X 0.00421
R_X 0.00421
R_X 0.00421
ZZ (−0.00313)
ZZ (−0.00313)
R_{XX} 1.28
ZZ (0.0875)
ZZ (0.0875)
R_Y $-\pi/2$
R_Y $-\pi/2$
R_Y $-\pi/2$
R_Y $-\pi/2$
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X 0.00421
3
2
1
0
R_X 0.00421
R_X 0.00421
R_X 0.00421
ZZ (−0.00313)
ZZ (−0.00313)
R_{XX} 1.28
ZZ (0.0875)
ZZ (0.0875)
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
X
X
X
X
12
13
14
15


q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.00421
ZZ (−0.00313)
ZZ (−0.00313)
R_{XX} 0.78
ZZ (0.587)
ZZ (0.587)
R_Y $-\pi/2$
|0⟩
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15
X

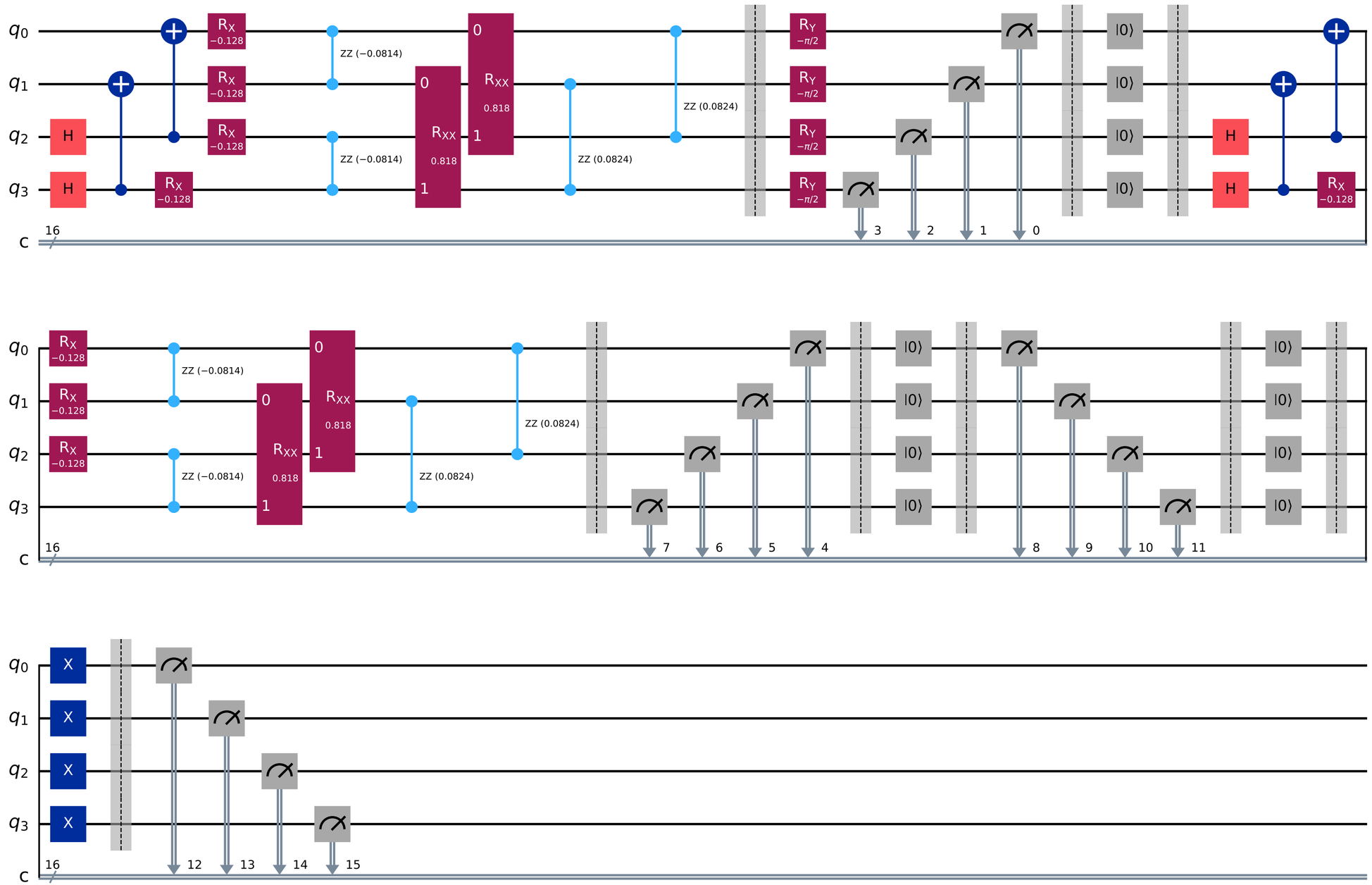

q0
q1
q2
q3
c
16
H
H
RX −0.311
RX −0.311
RX −0.311
RX −0.311
ZZ (−0.391)
ZZ (−0.391)
0
RXX 0.778
1
ZZ (0.0831)
ZZ (0.0831)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.311
RX −0.311
RX −0.311
ZZ (−0.391)
ZZ (−0.391)
0
RXX 0.778
1
ZZ (0.0831)
ZZ (0.0831)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
16
12
13
14
15

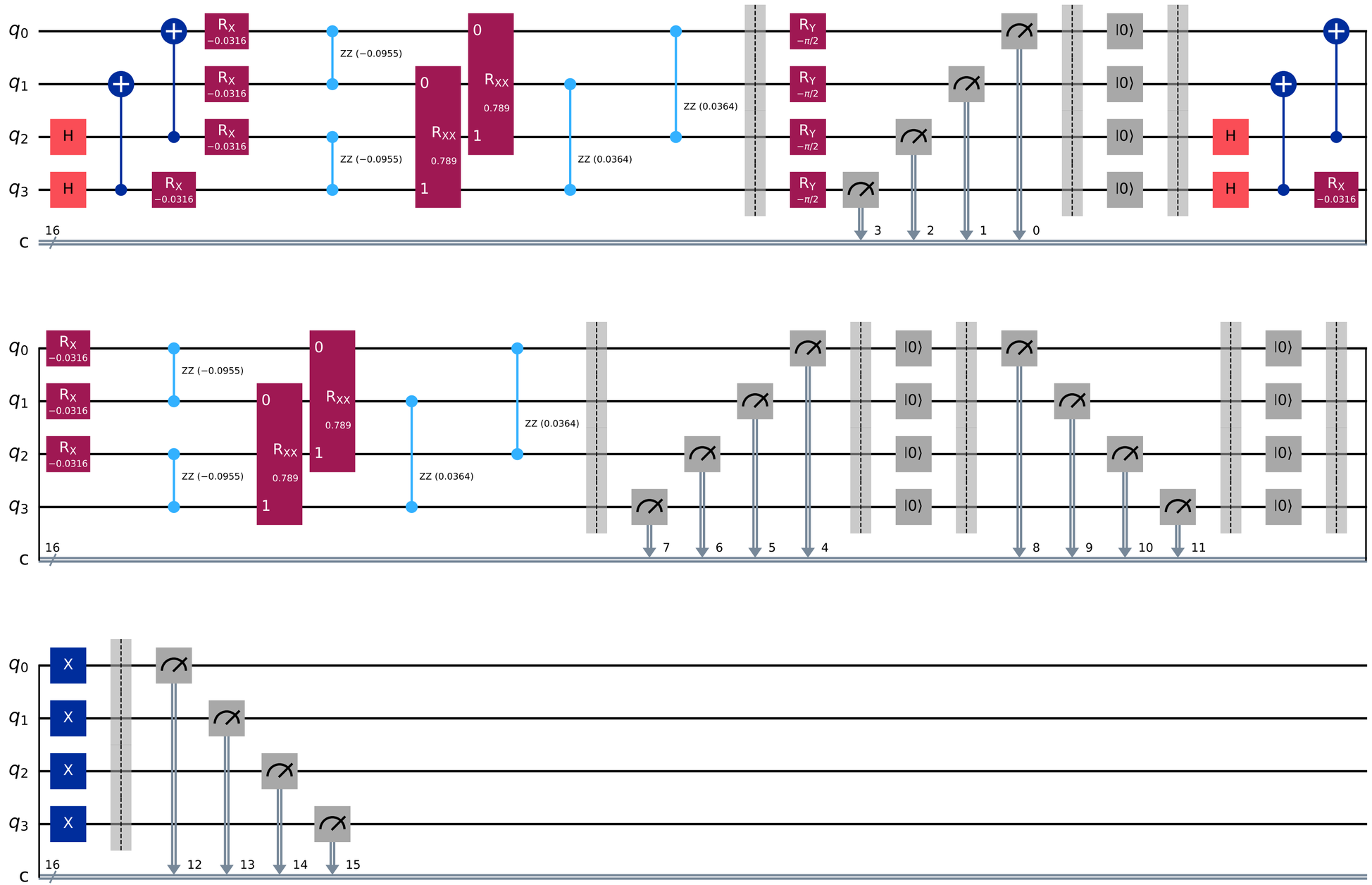

q0
q1
q2
q3
c
16
H
H
RX −0.125
RX −0.125
RX −0.125
RX −0.125
ZZ (0.211)
ZZ (0.211)
0
RXX 0.779
1
0
RXX 0.779
1
ZZ (0.0857)
ZZ (0.0857)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.125
RX −0.125
RX −0.125
RX −0.125
ZZ (0.211)
ZZ (0.211)
0
RXX 0.779
1
0
RXX 0.779
1
ZZ (0.0857)
ZZ (0.0857)
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

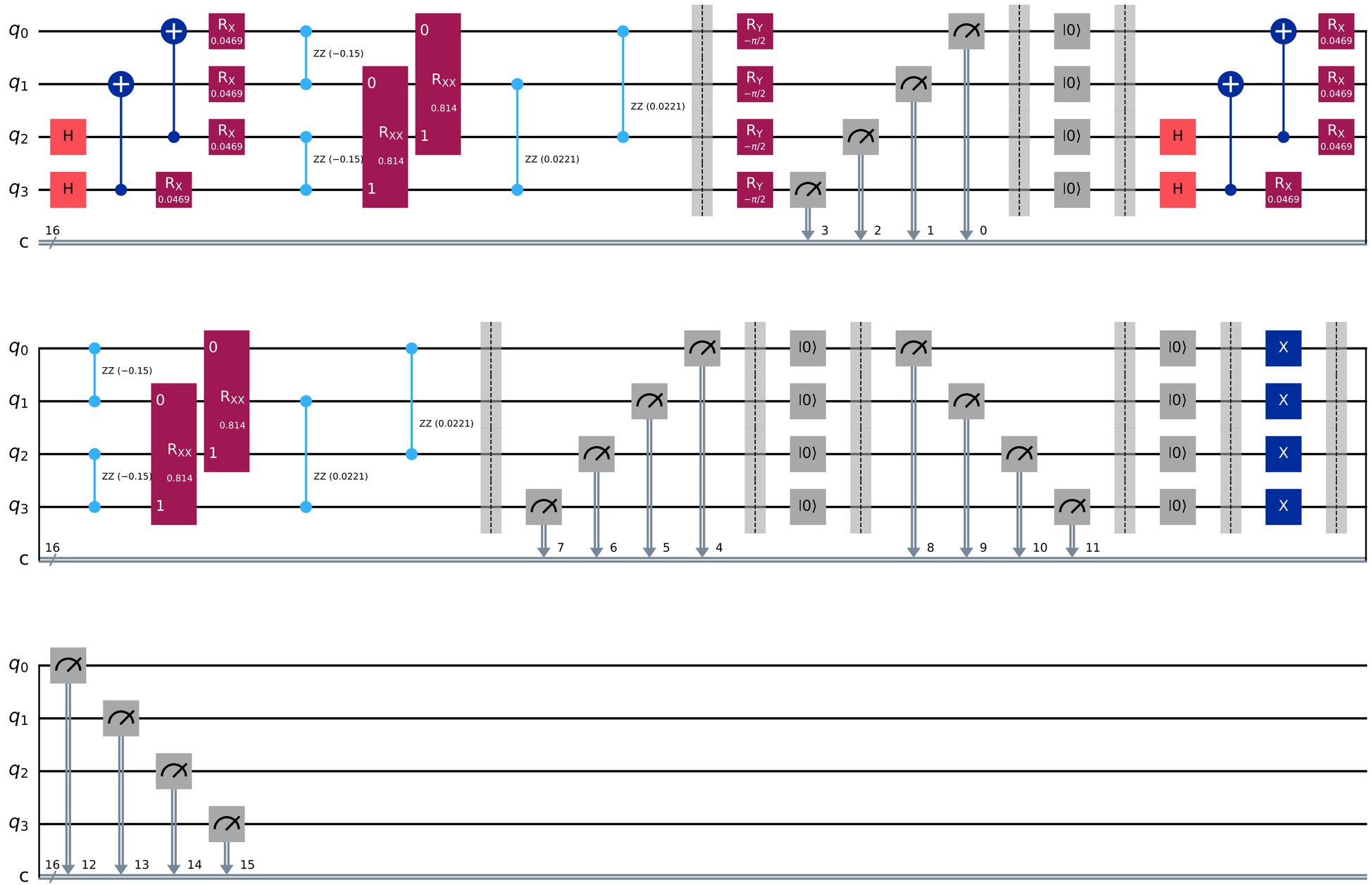

q0
q1
q2
q3
c
16
H
H
RX 0.103
RX 0.103
RX 0.103
RX 0.103
ZZ (−0.015)
ZZ (−0.015)
RXX 0.78
ZZ (0.0863)
ZZ (0.0863)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX 0.103
RX 0.103
RX 0.103
RX 0.103
ZZ (−0.015)
ZZ (−0.015)
RXX 0.78
ZZ (0.0863)
ZZ (0.0863)
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

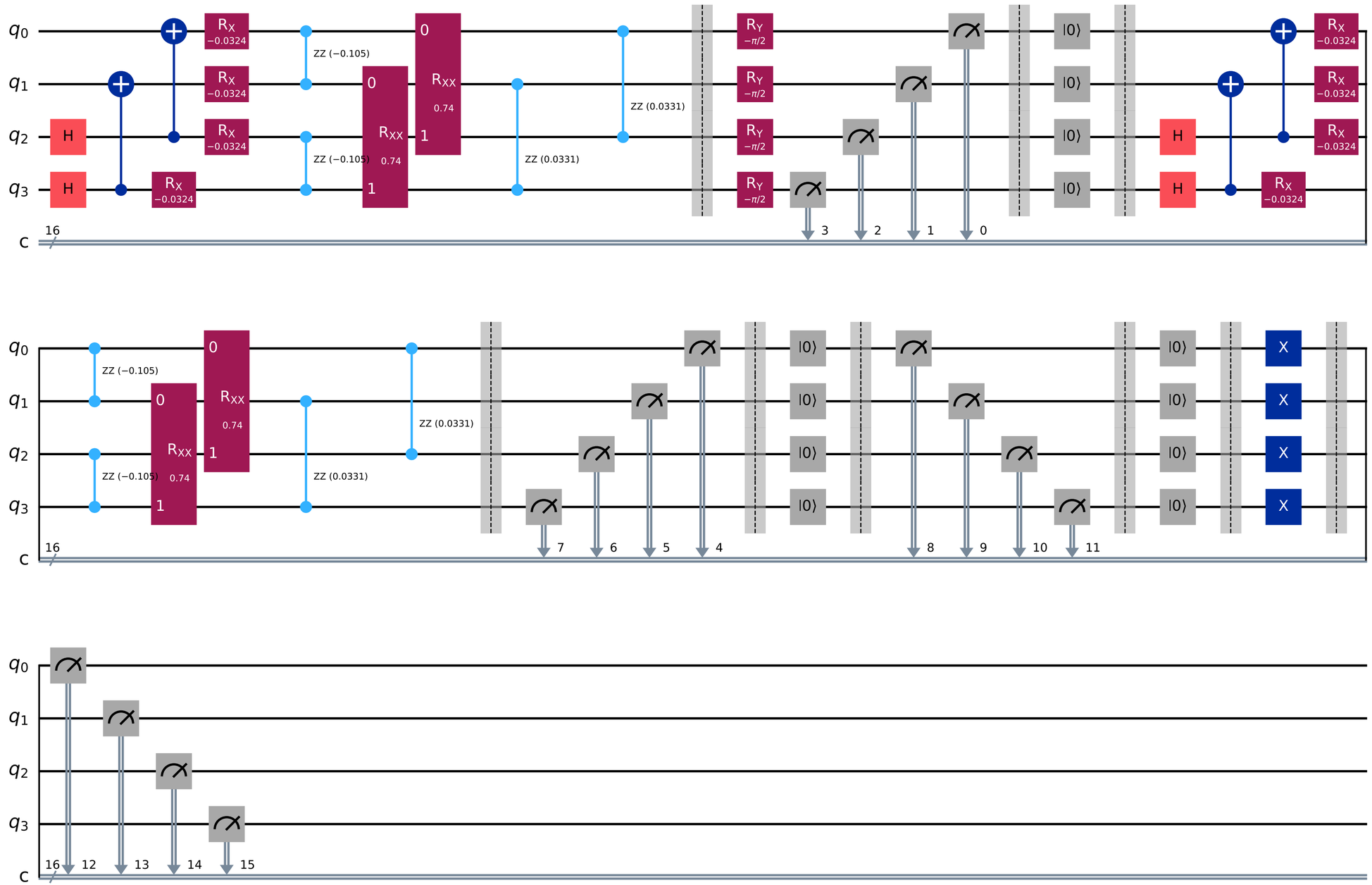

q0
q1
q2
q3
c
16
H
H
RX 0.0038
RX 0.0038
RX 0.0038
RX 0.0038
ZZ (−0.00363)
ZZ (−0.00363)
RXX 0.731
RXX 0.731
ZZ (0.0875)
ZZ (0.0875)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX 0.0038
3
2
1
0
RX 0.0038
RX 0.0038
RX 0.0038
ZZ (−0.00363)
ZZ (−0.00363)
RXX 0.731
RXX 0.731
ZZ (0.0875)
ZZ (0.0875)
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
X
X
X
X
12
13
14
15

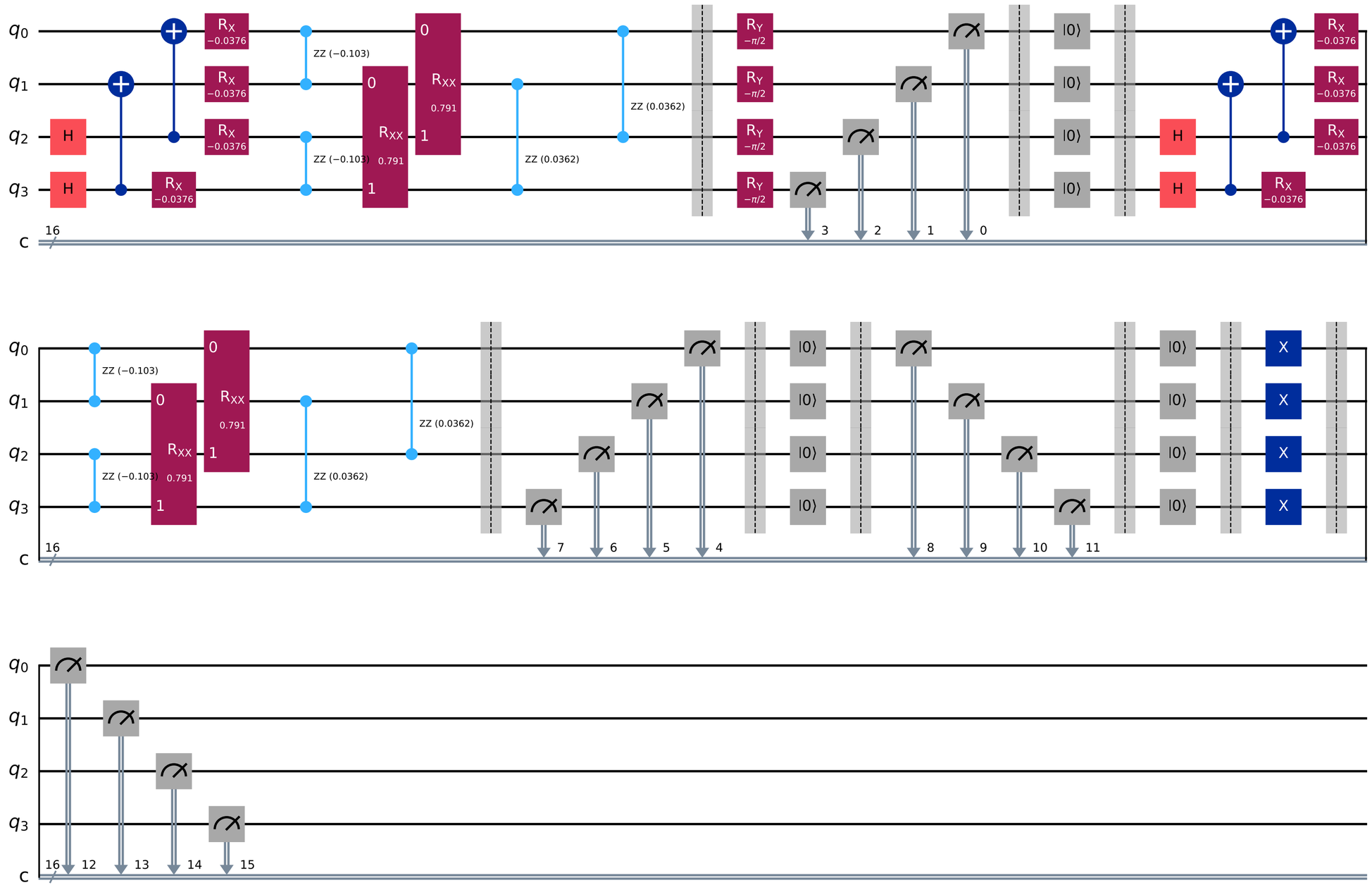
q0
q1
q2
q3
c
16
H
H
RX −0.0336
ZZ (−0.0953)
RXX 0.789
ZZ (0.0864)
RY −π/2
|0⟩
X
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15

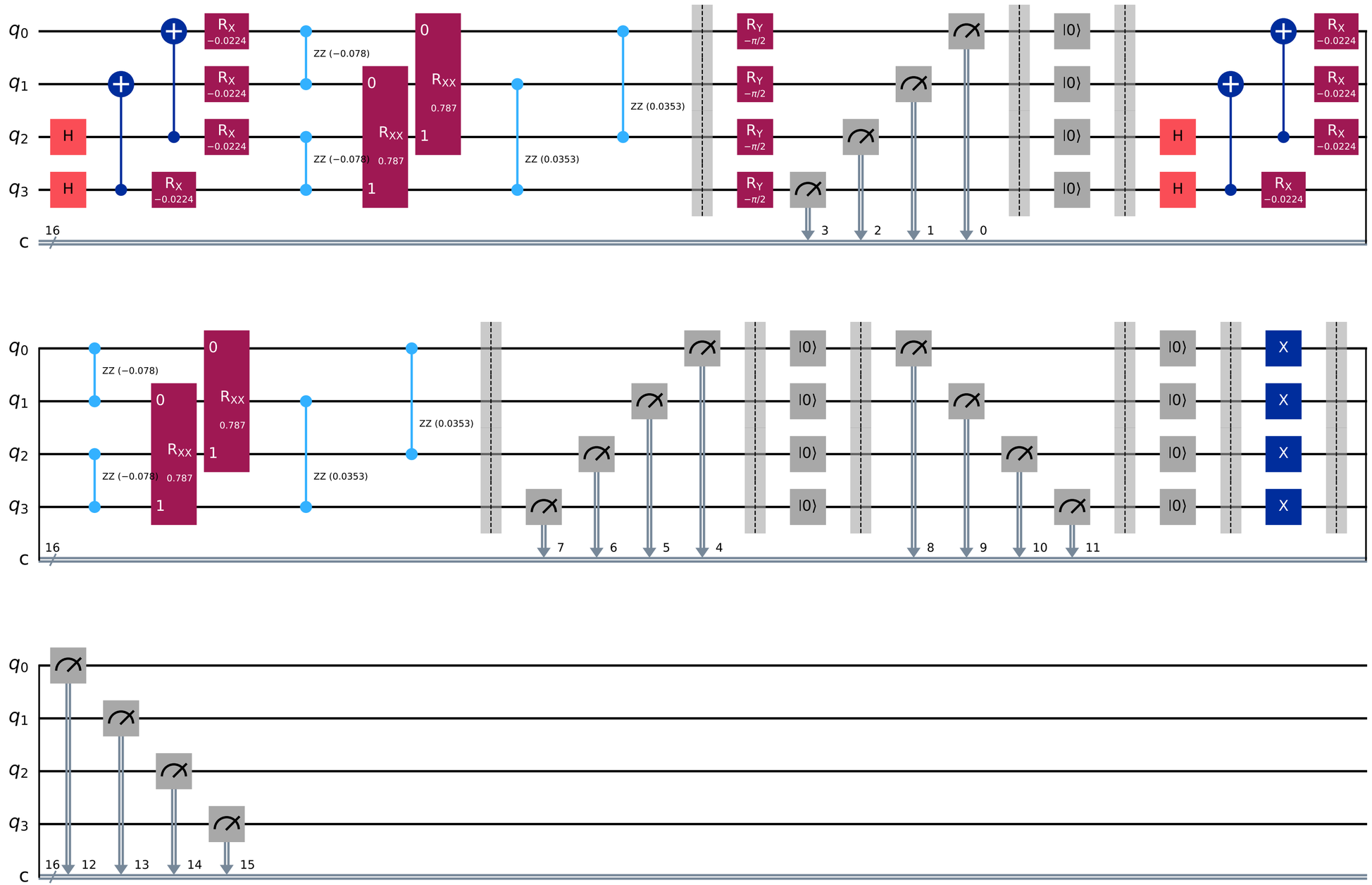

q0
q1
q2
q3
c
16
H
H
RX −0.128
RX −0.128
RX −0.128
RX −0.128
RX −0.128
ZZ (−0.0814)
ZZ (−0.0814)
0
RXX 0.818
1
0
RXX 0.818
1
ZZ (0.0824)
ZZ (0.0824)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.128
RX −0.128
RX −0.128
RX −0.128
ZZ (−0.0814)
ZZ (−0.0814)
0
RXX 0.818
1
0
RXX 0.818
1
ZZ (0.0824)
ZZ (0.0824)
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

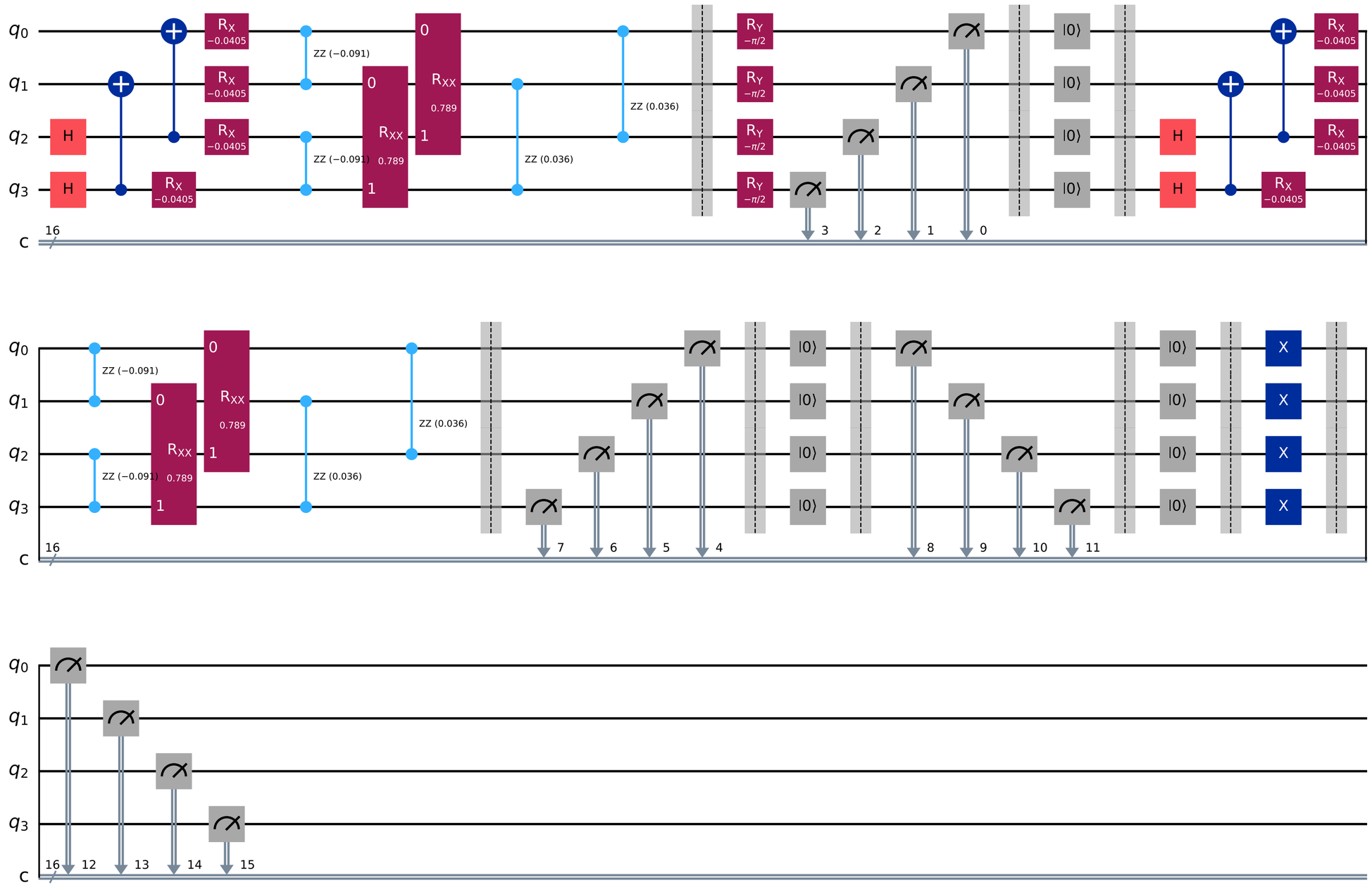

q0
q1
q2
q3
c
16
H
H
RX −0.0316
ZZ (−0.0955)
RXX 0.789
ZZ (0.0364)
RY −π/2
|0⟩
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15
X

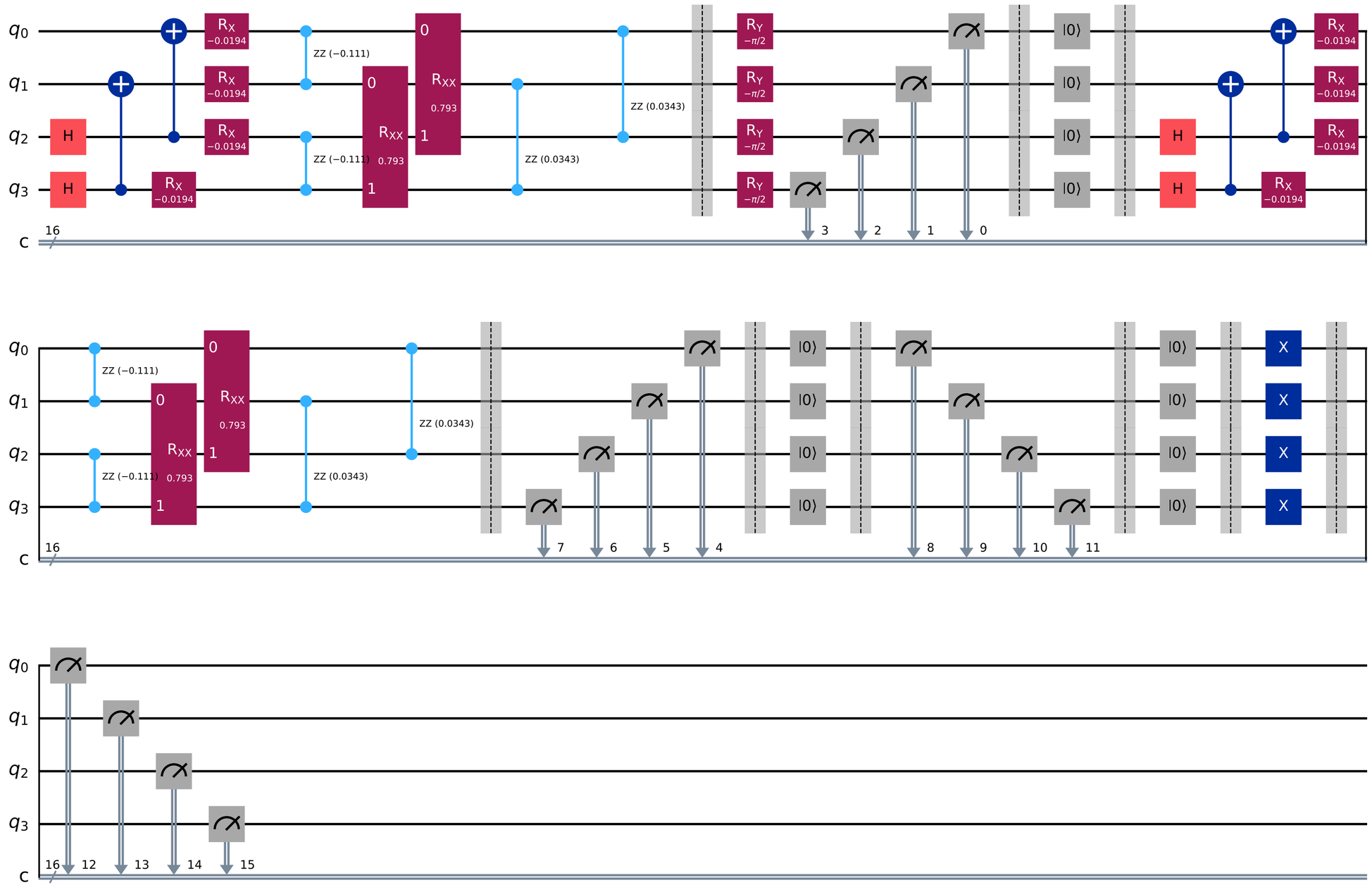

q_0
q_1
q_2
q_3
c
16
H
H
R_X 0.0469
R_{XX} 0.814
ZZ (−0.15)
ZZ (0.0221)
R_Y $-\pi/2$
|0⟩
X
3
2
1
0
7
6
5
4
8
9
10
11
12
13
14
15

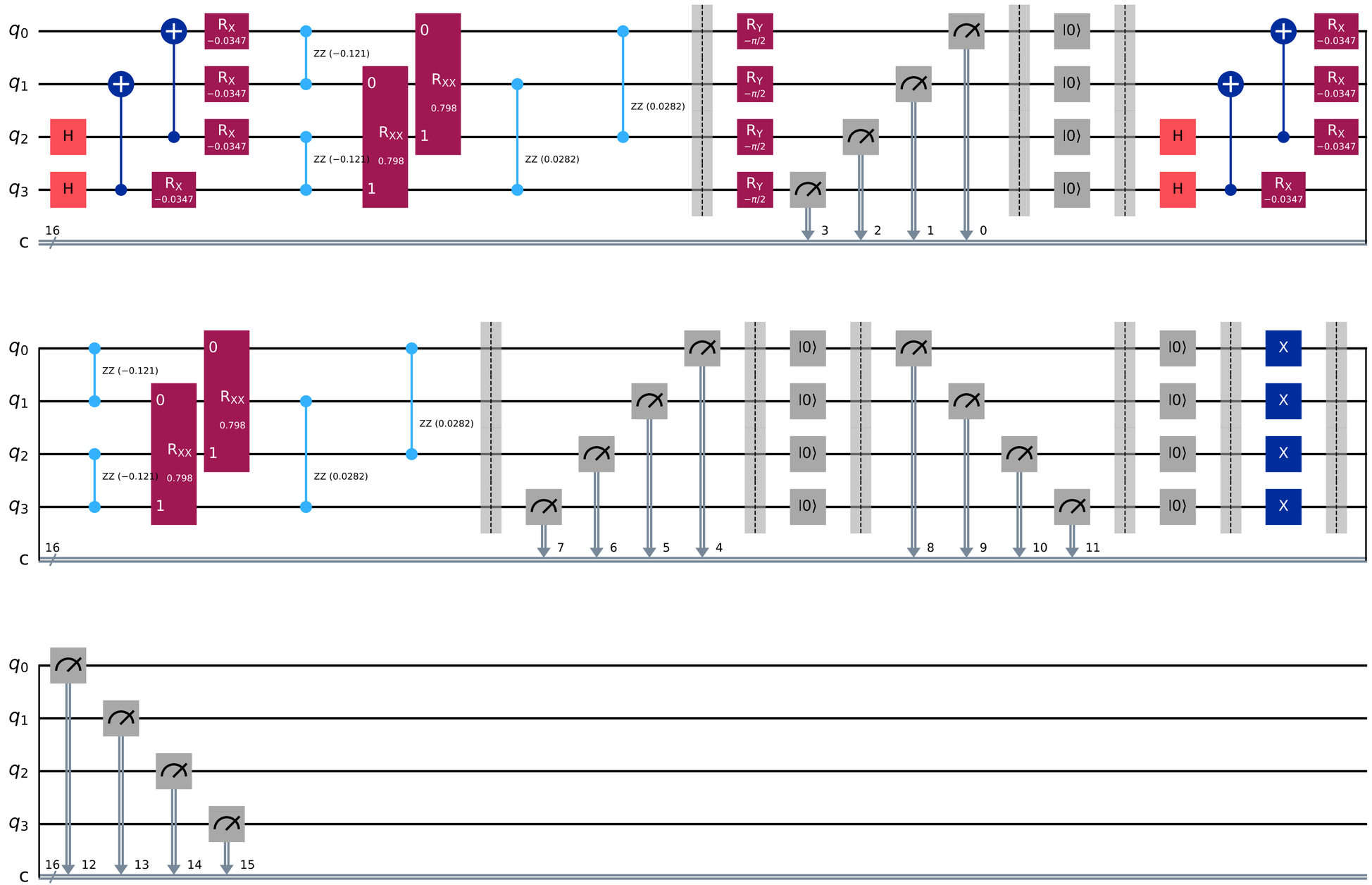

q0
q1
q2
q3
c
16
H
H
RX −0.0324
RX −0.0324
RX −0.0324
RX −0.0324
ZZ (−0.105)
ZZ (−0.105)
0
RXX 0.74
1
ZZ (0.0331)
ZZ (0.0331)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.0324
RX −0.0324
RX −0.0324
RX −0.0324
ZZ (−0.105)
ZZ (−0.105)
0
RXX 0.74
1
ZZ (0.0331)
ZZ (0.0331)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
16
12
13
14
15

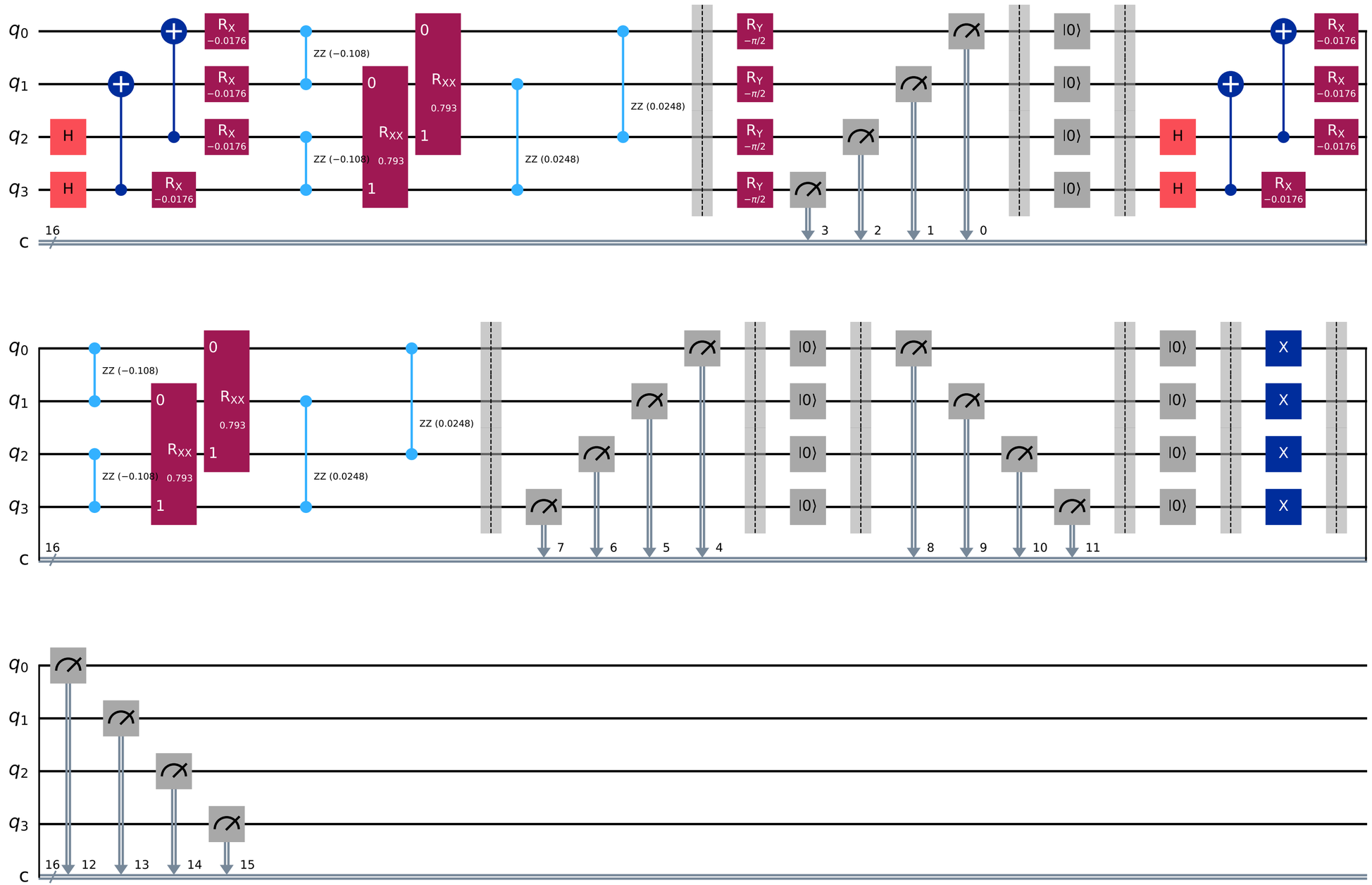

q0
q1
q2
q3
c
16
H
H
RX −0.0376
RX −0.0376
RX −0.0376
RX −0.0376
ZZ (−0.103)
ZZ (−0.103)
0
RXX 0.791
1
ZZ (0.0362)
ZZ (0.0362)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.0376
RX −0.0376
RX −0.0376
ZZ (−0.103)
ZZ (−0.103)
0
RXX 0.791
1
ZZ (0.0362)
ZZ (0.0362)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
16
12
13
14
15

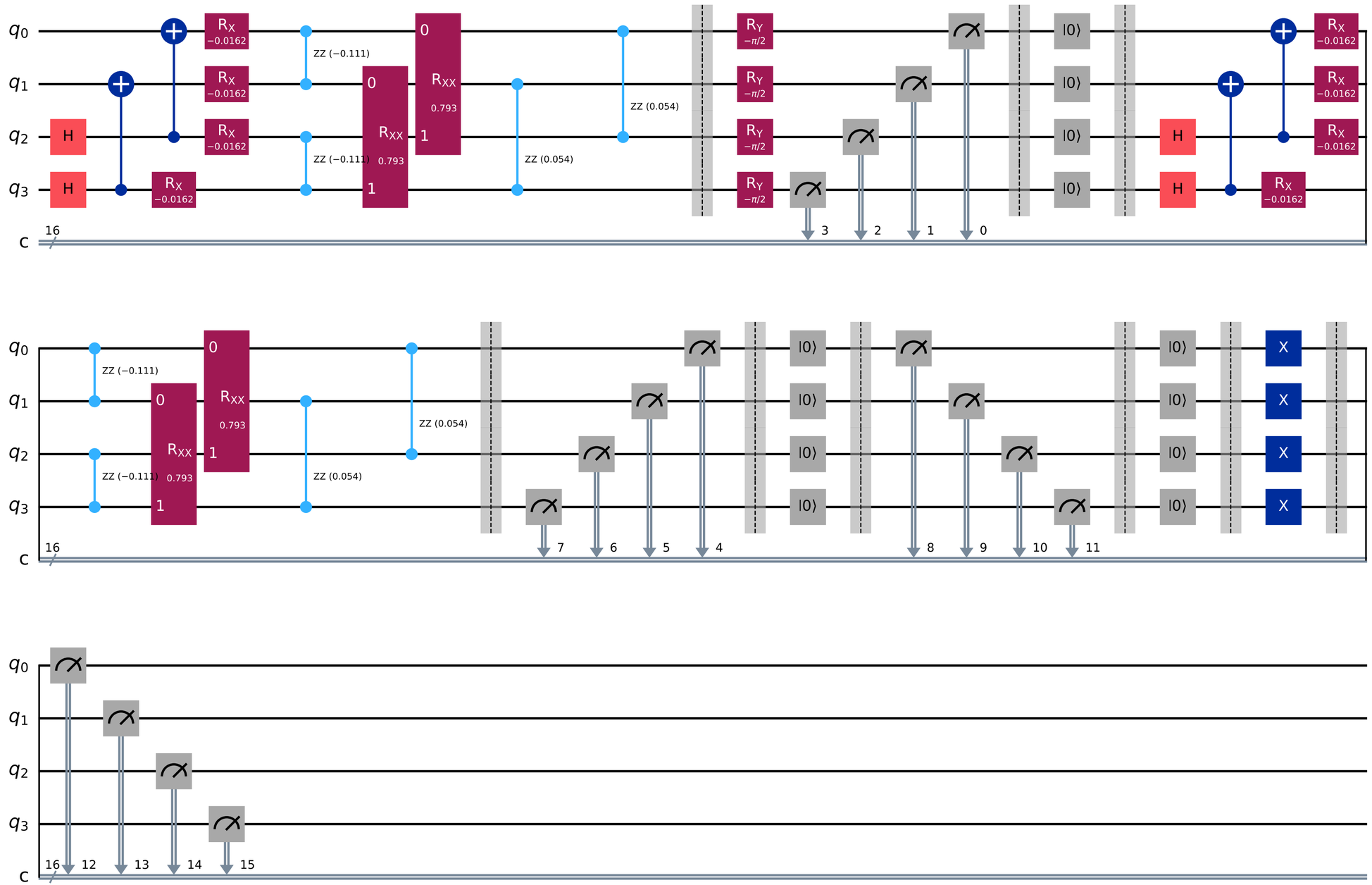

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.0224
R_X −0.0224
R_X −0.0224
R_X −0.0224
ZZ (−0.078)
ZZ (−0.078)
0
R_{XX}
0.787
1
0
R_{XX}
0.787
1
ZZ (0.0353)
ZZ (0.0353)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.0224
R_X −0.0224
R_X −0.0224
R_X −0.0224
ZZ (−0.078)
ZZ (−0.078)
0
R_{XX}
0.787
1
0
R_{XX}
0.787
1
ZZ (0.0353)
ZZ (0.0353)
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15

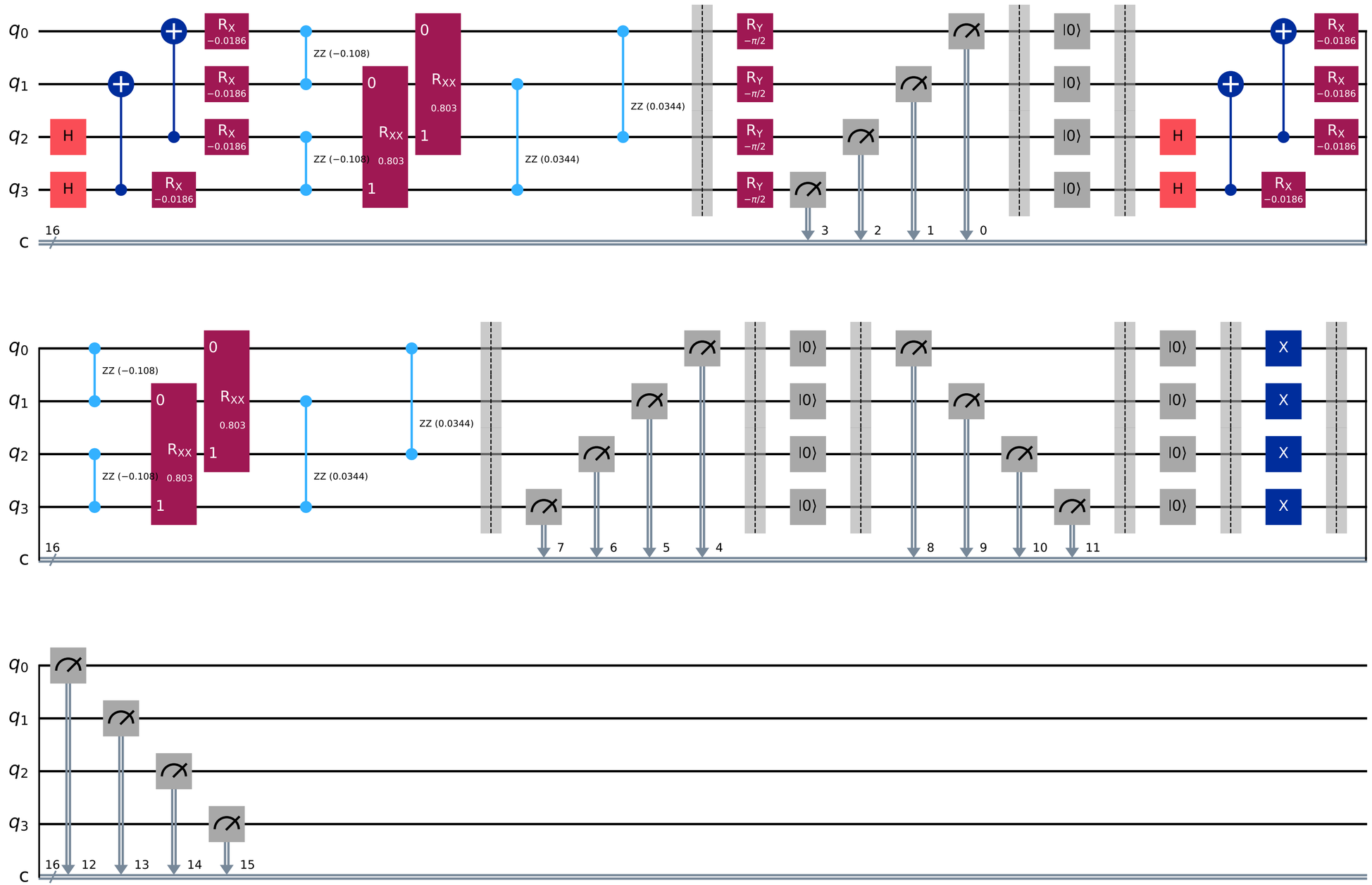

q0
q1
q2
q3
c
16
H
H
RX −0.0405
RX −0.0405
RX −0.0405
RX −0.0405
ZZ (−0.091)
ZZ (−0.091)
0
RXX
0.789
1
ZZ (0.036)
ZZ (0.036)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.0405
RX −0.0405
RX −0.0405
ZZ (−0.091)
ZZ (−0.091)
0
RXX
0.789
1
ZZ (0.036)
ZZ (0.036)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
16
12
13
14
15

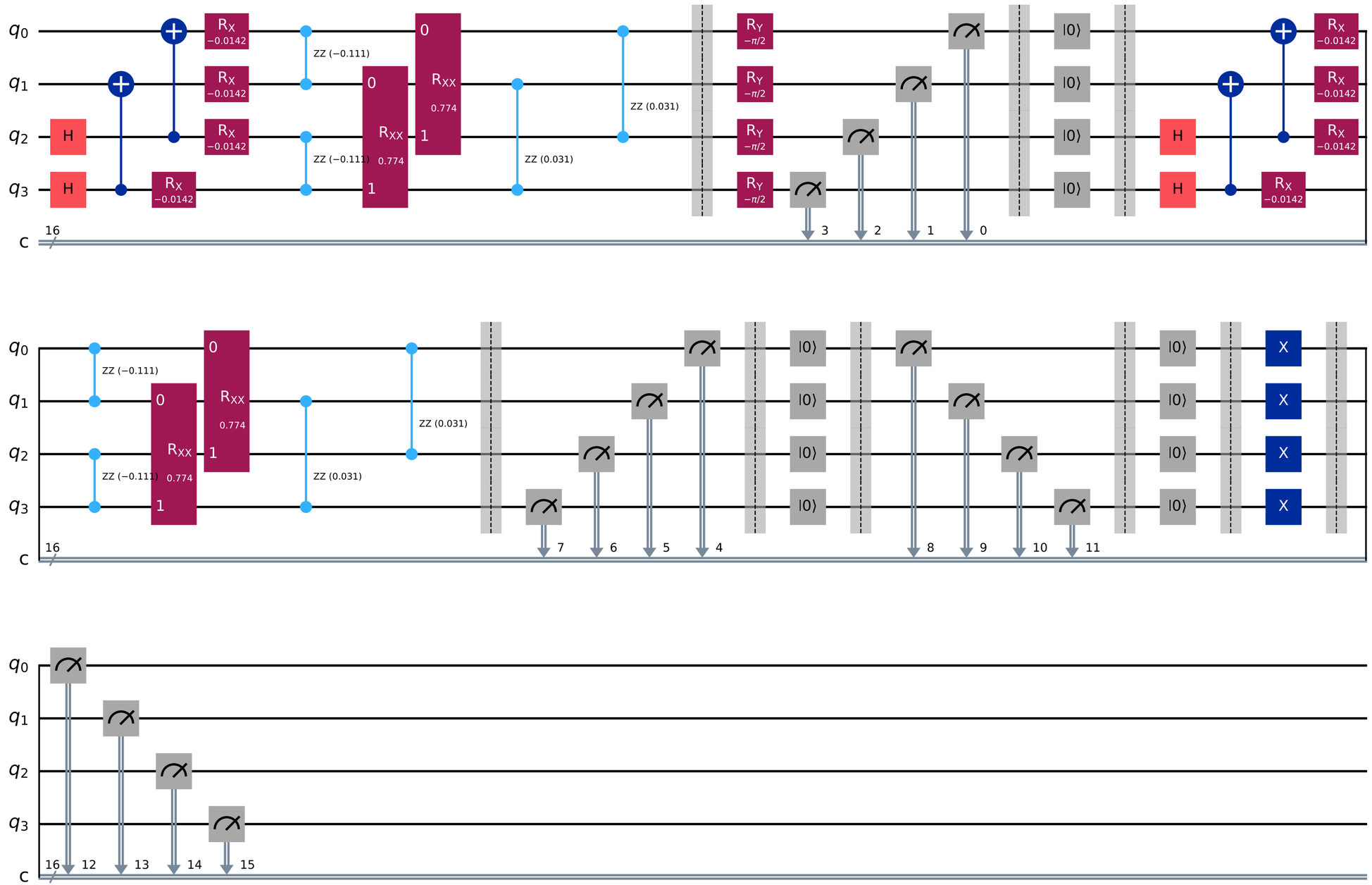

q0
q1
q2
q3
c
16
H
H
RX −0.0194
RX −0.0194
RX −0.0194
RX −0.0194
ZZ (−0.111)
ZZ (−0.111)
0
RXX
0.793
1
0
RXX
0.793
1
ZZ (0.0343)
ZZ (0.0343)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.0194
RX −0.0194
RX −0.0194
RX −0.0194
ZZ (−0.111)
ZZ (−0.111)
0
RXX
0.793
1
0
RXX
0.793
1
ZZ (0.0343)
ZZ (0.0343)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
16
12
13
14
15


q0
q1
q2
q3
c
16
H
H
RX −0.0347
RX −0.0347
RX −0.0347
RX −0.0347
ZZ (−0.121)
ZZ (−0.121)
0
RXX
0.798
1
ZZ (0.0282)
ZZ (0.0282)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.0347
RX −0.0347
RX −0.0347
RX −0.0347
ZZ (−0.121)
ZZ (−0.121)
0
RXX
0.798
1
ZZ (0.0282)
ZZ (0.0282)
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15


q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.0176
R_X −0.0176
R_X −0.0176
R_X −0.0176
ZZ (−0.108)
ZZ (−0.108)
0
R_{XX}
0.793
1
0
R_{XX}
0.793
1
ZZ (0.0248)
ZZ (0.0248)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.0176
R_X −0.0176
R_X −0.0176
R_X −0.0176
ZZ (−0.108)
ZZ (−0.108)
0
R_{XX}
0.793
1
0
R_{XX}
0.793
1
ZZ (0.0248)
ZZ (0.0248)
7
6
5
4
8
9
10
11
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
12
13
14
15


q0
q1
q2
q3
c
16
H
H
RX −0.0162
RX −0.0162
RX −0.0162
RX −0.0162
ZZ (−0.111)
ZZ (−0.111)
0
RXX 0.793
1
ZZ (0.054)
ZZ (0.054)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.0162
RX −0.0162
RX −0.0162
RX −0.0162
ZZ (−0.111)
ZZ (−0.111)
0
RXX 0.793
1
ZZ (0.054)
ZZ (0.054)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
12
13
14
15


q0
q1
q2
q3
c
16
H
H
RX −0.0186
RX −0.0186
RX −0.0186
RX −0.0186
ZZ (−0.108)
ZZ (−0.108)
0
RXX
0.803
1
0
RXX
0.803
1
ZZ (0.0344)
ZZ (0.0344)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.0186
RX −0.0186
RX −0.0186
RX −0.0186
ZZ (−0.108)
ZZ (−0.108)
0
RXX
0.803
1
0
RXX
0.803
1
ZZ (0.0344)
ZZ (0.0344)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
16
12
13
14
15


q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.0142
R_X −0.0142
R_X −0.0142
R_X −0.0142
ZZ (−0.111)
ZZ (−0.111)
R_{XX} 0.774
ZZ (0.031)
ZZ (0.031)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.0142
R_X −0.0142
R_X −0.0142
R_X −0.0142
ZZ (−0.111)
ZZ (−0.111)
R_{XX} 0.774
ZZ (0.031)
ZZ (0.031)
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
7
6
5
4
8
9
10
11
q_0
q_1
q_2
q_3
c
16
12
13
14
15